Trigonometric Identities and Integrals

Don't forget to add the constant of integration (C) to each indefinite integral!

Basic Trigonometric Integrals

These are the fundamental antiderivatives of the six core trigonometric functions:

- $\int \sin x \, dx = -\cos x + C$
- $\int \cos x \, dx = \sin x + C$
- $\int \tan x \, dx = -\ln|\cos x| + C = \ln|\sec x| + C$
- $\int \cot x \, dx = \ln|\sin x| + C$
- $\int \sec x \, dx = \ln|\sec x + \tan x| + C$
- $\int \csc x \, dx = -\ln|\csc x + \cot x| + C = \ln|\csc x - \cot x| + C$

Integrals of Trigonometric Products and Squares

These integrals frequently appear in calculus and physics, often derived using the reverse of the chain rule or standard trigonometric identities (like the half-angle or Pythagorean identities):

- $\int \sec^2 x \, dx = \tan x + C$
- $\int \csc^2 x \, dx = -\cot x + C$
- $\int \sec x \tan x \, dx = \sec x + C$
- $\int \csc x \cot x \, dx = -\csc x + C$
- $\int \sin^2 x \, dx = \frac{x}{2} - \frac{\sin(2x)}{4} + C$
- $\int \cos^2 x \, dx = \frac{x}{2} + \frac{\sin(2x)}{4} + C$
- $\int \tan^2 x \, dx = \tan x - x + C$
- $\int \cot^2 x \, dx = -\cot x - x + C$

Integrals Yielding Inverse Trigonometric Functions

When integrating certain rational and radical functions, the result is an inverse trigonometric function. In these formulas, a represents a positive constant ($a > 0$):

- $\displaystyle\int \frac{1}{\sqrt{a^2 - x^2}}\,dx = \arcsin\left(\frac{x}{a}\right) + C$
- $\displaystyle\int \frac{1}{a^2 + x^2}\,dx = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C$
- $\displaystyle\int \frac{1}{x\sqrt{x^2 - a^2}}\,dx = \frac{1}{a}\text{arcsec}\left(\frac{|x|}{a}\right) + C$

Trigonometric Substitutions for Integration

While not integrals themselves, these are the three core identity-based substitutions used to solve integrals containing specific radical forms (where $a > 0$):

If the integral contains...	**Use the substitution...**	**Based on the identity...**
$\sqrt{a^2 - x^2}$	$x = a\sin\theta$	$1 - \sin^2\theta = \cos^2\theta$
$\sqrt{a^2 + x^2}$	$x = a\tan\theta$	$1 + \tan^2\theta = \sec^2\theta$
$\sqrt{x^2 - a^2}$	$x = a\sec\theta$	$\sec^2\theta - 1 = \tan^2\theta$